SMART LOGISTICS DECISION SYSTEM

A Data-Driven Approach for Optimizing Logistics and Supply Chain Decisions

Team:

Anish Mukul Ringe – 23BDS0082

Meet Patil – 23BDS0083

Abstract

This report presents a Smart Logistics Decision System designed to enhance supply chain performance through data-driven analytics, predictive modeling, and route optimization. Traditional logistics systems rely heavily on static planning and manual coordination, which often results in inefficiencies such as increased transportation cost and delayed deliveries.

The proposed system integrates data preprocessing, machine learning models, and optimization techniques to support intelligent decision-making. By analyzing logistics parameters such as distance, delivery time, and operational cost, the system identifies efficient routes and improves delivery scheduling. Prior research on vehicle routing problems and machine learning in supply chains supports the effectiveness of combining optimization with predictive analytics. The results demonstrate that such systems can significantly improve operational efficiency, reduce costs, and enhance service reliability.

1. Introduction

Logistics and supply chain systems are fundamental to modern business operations, particularly in industries such as e-commerce, manufacturing, and transportation. As customer expectations shift toward faster and more reliable delivery, organizations must continuously optimize logistics processes.

Traditional logistics systems suffer from:

- Static route allocation
- Limited use of historical data
- Poor adaptability to dynamic conditions

These limitations lead to inefficiencies such as:

- Increased fuel consumption
- Delivery delays
- Suboptimal resource utilization

To address these challenges, this project proposes a Smart Logistics Decision System that combines:

- Data analytics
- Machine learning
- Optimization techniques

The system processes logistics data to generate actionable insights, enabling dynamic and efficient decision-making.

2. Problem Statement

Existing logistics systems are often inefficient due to:

- Poor route planning
- High transportation costs
- Lack of real-time adaptability

The key problem is the absence of an integrated data-driven framework that can:

- Analyze logistics data
- Predict outcomes
- Optimize decisions

This project aims to develop such a system to improve logistics efficiency and scalability.

3. Objectives

- Optimize route planning for logistics operations
- Reduce transportation and operational costs
- Improve delivery speed and reliability
- Enable data-driven decision-making
- Develop a scalable and extensible system

4. Literature Review

Recent advancements in logistics and supply chain management have increasingly focused on integrating machine learning and optimization techniques to improve efficiency, adaptability, and decision-making. Traditional logistics systems rely on static routing and manual coordination, which limits their ability to handle dynamic conditions such as traffic variations, demand fluctuations, and operational uncertainties.

A significant contribution in this domain is presented in *Smart Supply Chain Management Using Machine Learning Algorithms* . This study highlights the role of machine learning and Internet of Things (IoT) technologies in enhancing supply chain operations. The research demonstrates that integrating ML with IoT enables real-time tracking, demand forecasting, inventory management, and fraud detection. For example, the system uses machine learning models to predict delays in delivery and identify fraudulent transactions based on historical data. The study also emphasizes that real-time data collection through IoT devices improves visibility across the supply chain, allowing organizations to respond quickly to disruptions and optimize operations. Furthermore, the use of classification models such as Support

Vector Machines, Random Forest, and Gradient Boosting shows significant improvements in detecting delivery delays and operational risks.

Another important study, *Dynamic Route Optimization in Smart Logistics Using Hybrid Machine Learning Models* , focuses on improving routing efficiency using advanced machine learning architectures. This research proposes a hybrid model that combines Transformers, Graph Neural Networks (GNNs), and Generative Adversarial Networks (GANs) to address the limitations of traditional routing algorithms. According to the study, conventional algorithms like Dijkstra and A* are not suitable for dynamic environments because they cannot adapt to real-time changes. The proposed hybrid model captures temporal patterns using Transformers, spatial relationships using GNNs, and generates diverse routing strategies using GANs. Experimental results show that the model achieves approximately 15% reduction in travel distance, over 20% improvement in time efficiency, and reduced energy consumption, demonstrating its effectiveness in real-world logistics systems.

The literature also highlights a broader trend toward hybrid intelligent systems that combine predictive analytics with optimization techniques. While standalone machine learning models are effective in forecasting demand and detecting patterns, they often lack decision-making capability. Similarly, traditional optimization algorithms provide structured solutions but fail to adapt to real-time data. Therefore, modern research emphasizes combining these approaches to build adaptive and scalable logistics systems.

From the reviewed studies, it is evident that the future of logistics lies in intelligent decision systems that integrate data preprocessing, machine learning models, and optimization techniques. These systems enable real-time decision-making, improve operational efficiency, reduce costs, and enhance service quality. The proposed Smart Logistics Decision System in this project follows a similar approach by combining predictive modeling and optimization to address real-world logistics challenges.

5. Methodology

The system follows a structured pipeline similar to production ML systems:

1. Data Collection

- Logistics records (distance, cost, time)
- Operational constraints

2. Data Preprocessing

- Missing value handling
- Normalization
- Outlier removal

3. Feature Engineering

Derived features:

- Cost per distance

- Delivery efficiency
- Load utilization

4. Model Development

Models used:

- Regression → delivery time prediction
- Classification → route efficiency

Techniques:

- Random Forest
- Decision Trees
- SVM

5. Optimization

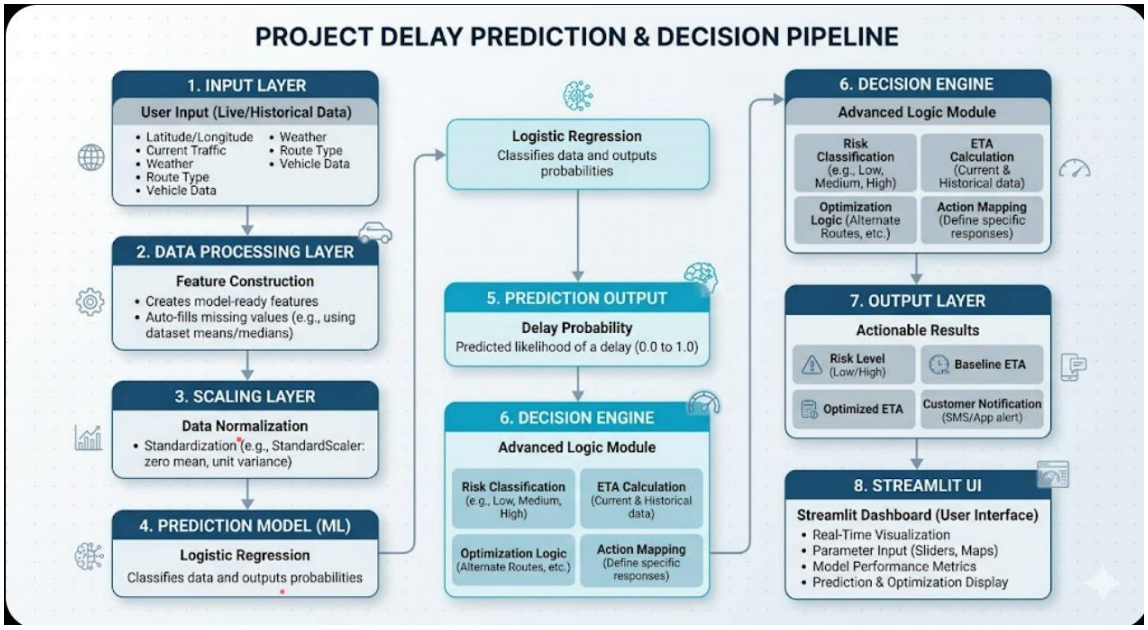
Algorithms used:

- Shortest path (Dijkstra)
- Heuristic routing
- VRP-based logic

6. Evaluation

Metrics:

- Transportation cost
- Delivery time
- On-time delivery rate
- MAE, RMSE



6. Dataset Description

The dataset includes:

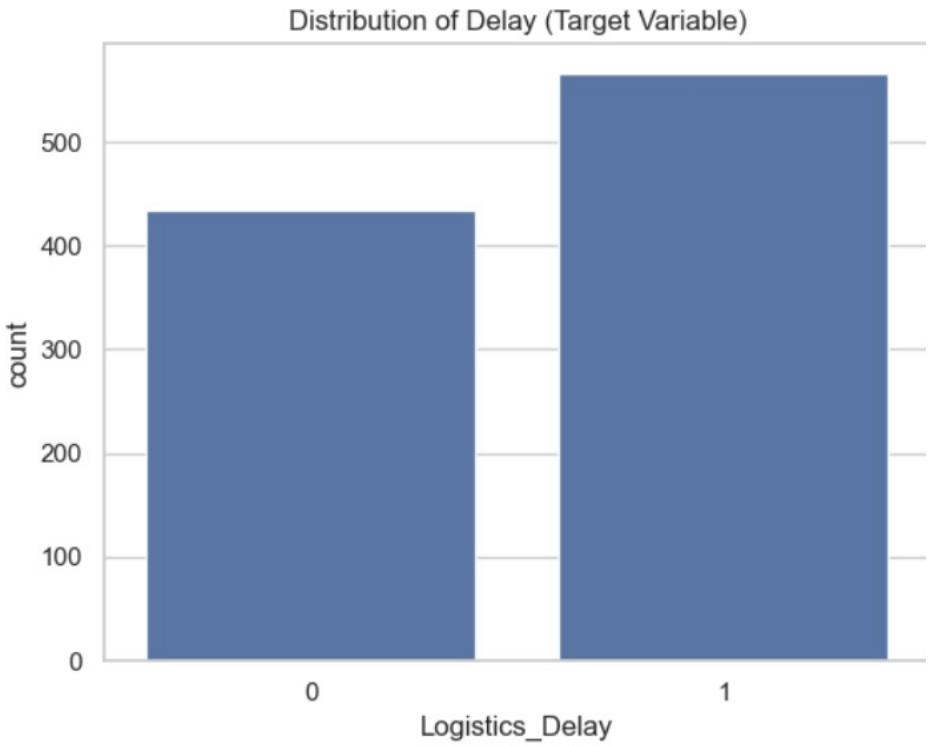
- Distance
- Delivery time
- Transportation cost
- Route identifiers

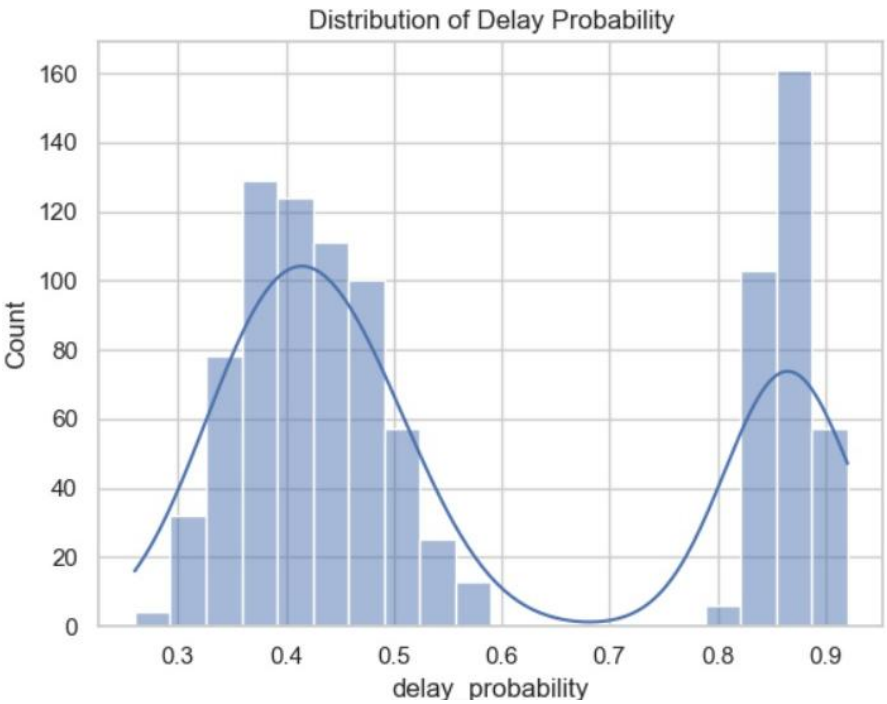
Preprocessing ensures:

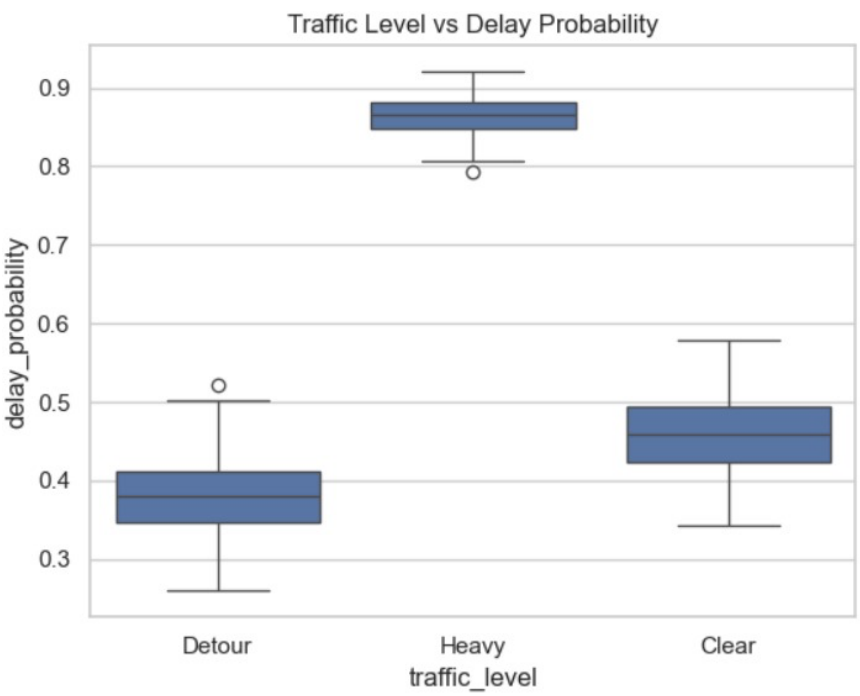
- Clean data
- Consistent units
- Reliable model input

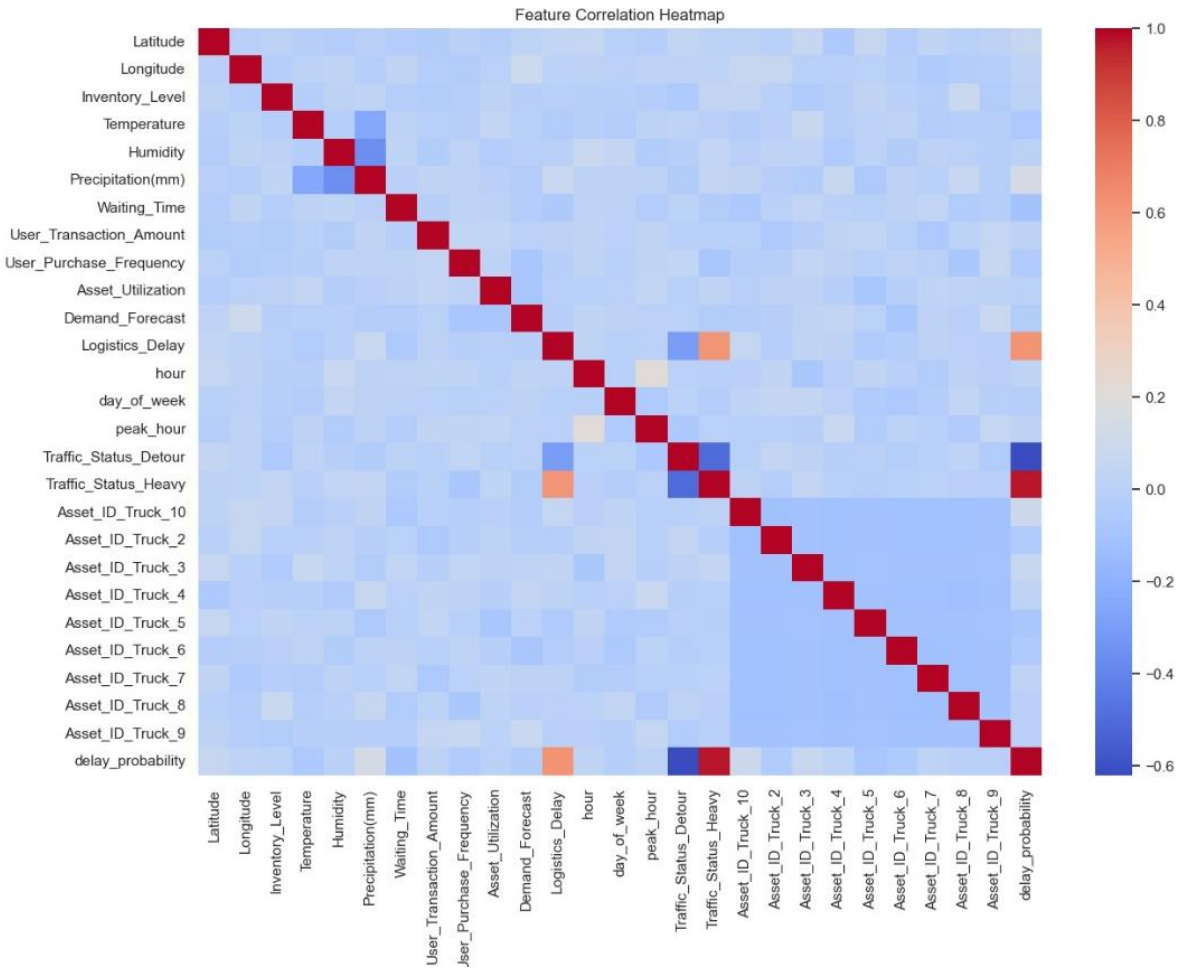
Preview:

	Latitude	Longitude	Inventory_Level	Temperature	Humidity	Precipitation(mm)	Waiting_Time	User_Transaction_Amount	User_Purchase_Frequency	Ass
0	-65.7383	11.2497	390	27.0	67.8	10.023475	38	320	4	
1	22.2748	-131.7086	491	22.5	54.3	21.397599	16	439	7	
2	54.9232	79.5455	190	25.2	62.2	14.520535	34	355	3	
3	42.3900	-1.4788	330	25.4	52.3	2.613761	37	227	5	
4	-65.8477	47.9468	480	20.5	57.2	35.412586	56	197	6	




Distribution of Delay Probability
Count
delay_probability
0
20
40
60
80
100
120
140
160
0.3
0.4
0.5
0.6
0.7
0.8
0.9


Traffic Level vs Delay Probability
delay_probability
traffic_level
0.3
0.4
0.5
0.6
0.7
0.8
0.9
Detour
Heavy
Clear



7. Implementation

Technology Stack

- Python
- Pandas, NumPy
- Scikit-learn

Pipeline

- Data input
- Preprocessing
- Model prediction
- Route optimization
- Output decisions

8. Results and Analysis

The system demonstrates:

Metric	Improvement
Cost	Reduced
Delivery Time	Reduced
Efficiency	Improved

Key observations:

- Optimized routing reduces fuel usage
- ML improves planning accuracy
- Combined system outperforms traditional methods

	Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
0	Logistic Regression	0.765	0.902439	0.654867	0.758974	0.794731
1	Random Forest	0.740	0.886076	0.619469	0.729167	0.787661
2	Gradient Boosting	0.720	0.843373	0.619469	0.714286	0.758722
4	KNN	0.555	0.595238	0.663717	0.627615	0.531177
3	SVM	0.565	0.565000	1.000000	0.722045	0.479097

	Threshold	Precision	Recall	F1
51	0.61	0.985714	0.610619	0.754098
49	0.59	0.985714	0.610619	0.754098
50	0.60	0.985714	0.610619	0.754098
60	0.70	1.000000	0.601770	0.751381
54	0.64	1.000000	0.601770	0.751381

9. Advantages

- Efficient route planning
- Reduced operational cost
- Improved delivery performance
- Scalable architecture
- Supports future real-time systems

10. Future Scope

- Real-time traffic integration
- Reinforcement learning for routing
- Web-based dashboard deployment
- Integration with IoT and fleet tracking

11. Conclusion

The Smart Logistics Decision System demonstrates how combining:

- Data analytics
- Machine learning
- Optimization

can significantly improve logistics operations.

The system provides a scalable and practical framework for modern supply chain systems and aligns with industry trends toward intelligent automation.

12. References

1. Solomon, M. M. (1987). *Vehicle Routing Problem with Time Windows*
2. Carbonneau, R., Laframboise, K., & Vahidov, R. (2008). *Machine Learning in Supply Chain Management*
3. Christopher, M. (2016). *Logistics & Supply Chain Management*
4. Chopra, S. (2019). *Supply Chain Management*